

Dyadic Spectral Determinants for the Fabius–Rvachev System

New zero-spectrum, local-jet, heat-trace, and q -binomial bridges

A frontier report based on the complete recursive `*.tex` corpus under

`Analysis/FabiusFunction/docs`

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Central outcome. The Fourier transform of Rvachev’s up-function is reorganized as the Fredholm determinant of a dyadically self-similar diagonal operator. This produces, in one framework, an exact spectral zeta function, a complex-dimension heat expansion, exact all-order jets at every integer zero, a multiscale q -Euler determinant, finite Gaussian-binomial coefficient formulae, and a symmetric q -integer deformation whose $q \uparrow 1$ limit is the original infinite sinc product.

Abstract

Let

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right), \quad \text{sinc } w = \frac{\sin w}{w}.$$

The inspected corpus already contains the canonical integer-zero product, the exact multiplicity $1 + \nu_2(m)$ at a nonzero integer m , and the Thue–Morse sign law between consecutive zeros. The present report does not relabel those facts as new. Instead it introduces the diagonal operator

$$De_{j,n} = 2^j n e_{j,n} \quad (j \geq 0, n \geq 1)$$

and proves

$$\Phi(z) = \det(I - z^2 D^{-2}), \quad D \simeq N \oplus 2D.$$

The decomposition gives the exact positive zero count $2M - s_2(M)$, the spectral zeta function

$$\zeta_D(s) = \frac{\zeta(s)}{1 - 2^{-s}},$$

and a heat trace with an explicit real-analytic log-periodic term. We derive a normalized, exact all-order Taylor factorization at every integer zero. We then define a dyadic q -Euler determinant

$$\mathcal{E}(u; Q) = \det(I - uQ^D) = \prod_{j \geq 0} (uQ^{2^j}; Q^{2^j})_{\infty},$$

obtain finite multiscale Gaussian-binomial expansions, and calculate the complete small- t Mellin asymptotics of $\mathcal{E}(u; e^{-t})$, including the critical $u = 1$ quadratic logarithm. Finally, with the symmetric q -integer $[m]_Q$, we construct

$$\Phi_Q(z) = \prod_{m \geq 1} \left(1 - \frac{z^2}{[m]_Q^2}\right)^{1 + \nu_2(m)}$$

and prove an exact q -Pochhammer factorization, the scaling law

$$\Phi_Q(z) = S_Q(z) \Phi_{Q^2}(z/[2]_Q),$$

and locally uniform convergence $\Phi_Q \rightarrow \Phi$ as $Q \uparrow 1$. This supplies a direct structural bridge from the Fourier zero divisor to q -Pochhammer symbols and q -binomial coefficients.

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1 Scope, corpus audit, and novelty protocol

1.1 What was inspected

The recursive documentation tree at <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs> is not a single article. It is a stratified mathematical record containing living expositions, an explicitly frozen historical archive, a consolidated non-formalized frontier volume, a Thue–Morse formula atlas, twelve successive Fourier-decay analyses, and imported source transcriptions of external papers. The corpus ledger in [section A](#) records 57 TeX artifacts:

Stratum	TeX artifacts	Role
Living and canonical documents	18	Current expositions and active frontier drafts
Frozen historical archive	32	Provenance, superseded drafts, and deltas
Imported paper sources	7	External mathematical context
Total	57	Recursive corpus audited

Every TeX artifact was included in the audit. Mathematical extraction was concentrated on the living/canonical descendants and the active Fourier series; frozen ancestors were compared through their archived first-version source, provenance descriptions, and surviving distinctive sections. This distinction matters: the archive intentionally preserves superseded states and explicitly prunes byte-identical copies. Treating all 57 files as independent contemporary claims would create false novelty and heavy duplication.

1.2 Three status classes

Status convention. Throughout this report, *established baseline* means a statement already present in the inspected repository or in a cited external source. *New deduction* means that the theorem and proof were not found in the inspected corpus or in targeted searches made for this report. This is a relative corpus claim, not a universal priority claim. The proofs below are mathematical proofs written for human verification; they have not yet been checked by Lean.

The main examples of statements that are *not* claimed here as new are:

- (i) the infinite sinc product for $\widehat{\text{up}}$ and its dyadic scaling equation;
- (ii) the canonical product
$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)};$$
- (iii) the assertion that the nonzero zeros are precisely the integers, with multiplicity $1 + \nu_2(|m|)$;
- (iv) strict logarithmic concavity of $|\Phi|$ between consecutive zeros and the sign formula
$$\text{sgn } \Phi(x) = (-1)^{s_2(M)} \quad (M < x < M + 1);$$
- (v) the existing moment/cumulant formulae for the Rvachev probability law;
- (vi) the base-1/2 q -Pochhammer and Gaussian-binomial formulae for exact dyadic values of the Fabius function;
- (vii) the pointwise, metric, mean, extremal, and annular Fourier-decay results assembled in the final Fourier-decay synthesis.

The purpose of the report is instead to place those facts inside a new spectral determinant and q -deformation architecture and to extract consequences not present in the source corpus.

1.3 Result map

New deductions developed below. The report proves: (a) a Fredholm determinant model with $D \simeq N \oplus 2D$; (b) exact zero-counting and spectral-parity formulae; (c) the meromorphic spectral zeta $\zeta(s)/(1 - 2^{-s})$ and its heat trace with explicit log-periodic oscillation; (d) exact normalized all-order jets at every integer zero; (e) an operator-valued dyadic q -Euler product and finite multiscale Gaussian-binomial expansion; (f) complete small- t asymptotics of the resulting tower of q -Pochhammer symbols; (g) a symmetric q -integer deformation of Φ with exact q -scaling, zero arithmetic, and $q \uparrow 1$ convergence; and (h) a base- b extension explaining which features are genuinely dyadic.

2 Established baseline and notation

2.1 The twin functions

Rvachev's up-function up is the even C^∞ density supported on $[-1, 1]$ whose Fourier transform, under the convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i z x} dx,$$

is

$$\Phi(z) := \widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right). \quad (2.1)$$

The Fabius function F is its affine cumulative twin; on the support one may write

$$\text{up}(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1. \end{cases}$$

For the present report the Fourier entire function Φ is the central object, while the relation to F explains why the resulting identities belong to the same theory.

A probability realization is useful. If $U_0, U_1, \dots$ are independent and uniform on $[-1/2, 1/2]$, then

$$X = \sum_{j=0}^{\infty} 2^{-j} U_j \quad (2.2)$$

has density up and characteristic function $\mathbb{E} e^{itX} = \Phi(t/(2\pi))$.

2.2 Thue–Morse and binary notation

For $n \geq 0$, let $s_2(n)$ denote the sum of the binary digits of n , and put

$$\tau_n = (-1)^{s_2(n)}. \quad (2.3)$$

The sequence satisfies $\tau_{2n} = \tau_n$ and $\tau_{2n+1} = -\tau_n$. We write $\nu_2(m)$ for the exponent of 2 in a positive integer m .

2.3 q -notation

For $|Q| < 1$,

$$(a; Q)_L = \prod_{r=0}^{L-1} (1 - aQ^r), \quad (a; Q)_\infty = \prod_{r=0}^{\infty} (1 - aQ^r),$$

and the Gaussian binomial coefficient is

$$\begin{bmatrix} L \\ k \end{bmatrix}_Q = \frac{(Q; Q)_L}{(Q; Q)_k (Q; Q)_{L-k}}.$$

The finite q -binomial theorem is

$$(a; Q)_L = \sum_{k=0}^L (-1)^k Q^{\binom{k}{2}} \begin{bmatrix} L \\ k \end{bmatrix}_Q a^k. \quad (2.4)$$

The corpus uses these functions chiefly at base $Q = 1/2$ to encode exact dyadic values of F . The constructions below use a different Q : it deforms the *Fourier zero spectrum*. The two roles should not be conflated.

2.4 The established zero divisor

Euler's sine product applied scale by scale to equation (2.1) gives

$$\Phi(z) = \prod_{j=0}^{\infty} \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{(2^j n)^2} \right) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2} \right)^{a_m}, \quad a_m := 1 + \nu_2(m). \quad (2.5)$$

The second equality counts the representations $m = 2^j n$. Consequently the nonzero zeros are exactly $\pm m$, each with multiplicity a_m . The active Fourier-decay corpus already contains equation (2.5); it is the starting point, not a new theorem of this report.

3 The dyadic spectral operator

3.1 Definition and self-similarity

Let

$$\mathcal{H} = \bigoplus_{j=0}^{\infty} \ell^2(\mathbb{N}),$$

with orthonormal basis $e_{j,n}$, $j \geq 0$, $n \geq 1$. Define the positive diagonal operator

$$D e_{j,n} = 2^j n e_{j,n}. \quad (3.1)$$

Let N be the number operator on $\ell^2(\mathbb{N})$, $N e_n = n e_n$.

Theorem 3.1 (Dyadic spectral decomposition). *There is a unitary equivalence*

$$D \simeq N \oplus 2D. \quad (3.2)$$

Moreover D^{-p} is trace class precisely for $p > 1$.

Proof. The subspace with $j = 0$ carries the eigenvalues n , hence is N . Relabeling $e_{j,n}$ as $e_{j-1,n}$ on the subspace $j \geq 1$ identifies the restriction with $2D$. Finally,

$$\mathrm{Tr} D^{-p} = \sum_{j=0}^{\infty} \sum_{n=1}^{\infty} (2^j n)^{-p} = \frac{\zeta(p)}{1 - 2^{-p}},$$

which is finite exactly when $p > 1$. □

The decomposition is the operator-theoretic form of dyadic refinement. It is also the source of every denominator $1 - 2^{-s}$ below.

3.2 The sinc product as a Fredholm determinant

Since D^{-2} is trace class, the ordinary Fredholm determinant $\det(I - z^2 D^{-2})$ is defined for every $z \in \mathbb{C}$.

Theorem 3.2 (Fredholm determinant representation). *For every $z \in \mathbb{C}$,*

$$\boxed{\Phi(z) = \det(I - z^2 D^{-2}).} \quad (3.3)$$

The operator decomposition equation (3.2) implies the exact scaling equation

$$\Phi(z) = \text{sinc}(\pi z) \Phi(z/2). \quad (3.4)$$

Proof. The eigenvalues of D^{-2} are $(2^j n)^{-2}$, and their sum is finite. Thus

$$\det(I - z^2 D^{-2}) = \prod_{j=0}^{\infty} \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{(2^j n)^2}\right).$$

Euler's product $\text{sinc}(\pi w) = \prod_{n \geq 1} (1 - w^2/n^2)$ identifies this with equation (2.1). Applying the multiplicativity of the determinant to $D \simeq N \oplus 2D$ gives

$$\det(I - z^2 D^{-2}) = \det(I - z^2 N^{-2}) \det(I - z^2 (2D)^{-2}) = \text{sinc}(\pi z) \Phi(z/2).$$

□

Remark 3.3. This is an *ordinary* Fredholm determinant, not a zeta-regularized determinant. The trace-class input is D^{-2} , and no renormalization is required.

3.3 Multiplicity as spectral degeneracy

The eigenvalue m of D occurs once for every j such that $2^j \mid m$. Therefore

$$\text{mult}_D(m) = a_m = 1 + \nu_2(m), \quad a_{2m} = a_m + 1, \quad a_{2m+1} = 1. \quad (3.5)$$

Thus the canonical zero multiplicity is exactly the spectral degeneracy of D . The elementary 2-regular recursion in equation (3.5) is the discrete shadow of $D \simeq N \oplus 2D$.

4 Exact zero arithmetic and spectral parity

4.1 The positive counting function

Let

$$A(M) = \text{rank } \mathbf{1}_{[1,M]}(D) = \sum_{m=1}^M a_m$$

count positive eigenvalues, including multiplicity.

Theorem 4.1 (Exact Weyl law with binary remainder). *For every integer $M \geq 0$,*

$$\boxed{A(M) = \sum_{j=0}^{\infty} \left\lfloor \frac{M}{2^j} \right\rfloor = 2M - s_2(M).} \quad (4.1)$$

Consequently, with $M = \lfloor R \rfloor$, the number of nonzero zeros of Φ in $|z| \leq R$, counted with multiplicity, is

$$\boxed{\mathcal{N}_{\Phi}(R) = 4M - 2s_2(M).} \quad (4.2)$$

Proof. Counting the pairs (j, n) with $2^j n \leq M$ gives the floor sum. On the other hand,

$$A(M) = M + \sum_{m=1}^M \nu_2(m) = M + \nu_2(M!).$$

Legendre's binary formula $\nu_2(M!) = M - s_2(M)$ proves equation (4.1). The zeros are symmetric, so the disk count is twice $A(M)$. $\square$

The exact formula refines the linear Weyl term $A(M) \sim 2M$ by a digit-sum fluctuation. In particular,

$$2M - \lfloor \log_2 M \rfloor - 1 \leq A(M) \leq 2M - 1 \quad (M \geq 1). \quad (4.3)$$

The recursions

$$A(2M) = A(M) + 2M, \quad A(2M + 1) = A(M) + 2M + 1 \quad (4.4)$$

follow immediately.

4.2 Thue–Morse as spectral-flow parity

Corollary 4.2 (Spectral parity formula). *For every $M \geq 0$,*

$$\boxed{(-1)^{\text{rank } \mathbf{1}_{[1, M]}(D)} = (-1)^{A(M)} = (-1)^{s_2(M)} = \tau_M.} \quad (4.5)$$

Hence, for $M < x < M + 1$,

$$\text{sgn } \Phi(x) = (-1)^{\text{rank } \mathbf{1}_{[1, M]}(D)} = \tau_M.$$

Proof. The even term $2M$ disappears modulo 2 in equation (4.1). Starting with $\Phi(0) = 1$, crossing the zeros $1, \dots, M$ multiplies the sign by $(-1)^{A(M)}$. $\square$

The interval-sign theorem was already present in the Fourier corpus. What is new here is its interpretation: the Thue–Morse bit is the parity of a finite-rank spectral projection of a self-similar operator.

4.3 Dyadic-block statistics

For uniform M in $\{0, \dots, 2^n - 1\}$, the random variable $s_2(M)$ is binomial $\text{Bin}(n, 1/2)$. Thus

$$\frac{1}{2^n} \sum_{M=0}^{2^n-1} (2M - A(M)) = \frac{n}{2}, \quad \text{Var}(2M - A(M)) = \frac{n}{4}. \quad (4.6)$$

The de Moivre–Laplace theorem gives

$$\frac{A(M) - (2M - n/2)}{\sqrt{n}/2} \implies \mathcal{N}(0, 1) \quad (4.7)$$

under dyadic-block sampling. Thus the deterministic zero-count remainder has an exact probabilistic law when sampled on its natural self-similar blocks.

4.4 A certified canonical-product truncation

Let

$$\Phi_M(z) = \prod_{m=1}^M \left(1 - \frac{z^2}{m^2}\right)^{a_m}.$$

Proposition 4.3 (Elementary tail bound). *For $M \geq 1$,*

$$T_M := \sum_{m>M} \frac{a_m}{m^2} < \frac{2}{M} + \frac{2\zeta(2)}{M^2}. \quad (4.8)$$

If $|z| \leq R$ and $M \geq \sqrt{2}R$, then, on any zero-free simply connected region where the logarithm is chosen continuously,

$$\left| \log \frac{\Phi(z)}{\Phi_M(z)} \right| \leq 2R^2 T_M. \quad (4.9)$$

Proof. Write the tail spectrally:

$$T_M = \sum_{j=0}^{\infty} 4^{-j} \sum_{n>M/2^j} \frac{1}{n^2}.$$

For $j \leq \lfloor \log_2 M \rfloor$, the inner tail is at most $2^j/M$; summing gives less than $2/M$. For the remaining j , bound the inner sum by $\zeta(2)$ and the geometric tail by $2/M^2$. If $|z|^2/m^2 \leq 1/2$, then $|\log(1-w)| \leq 2|w|$, and summation proves equation (4.9). $\square$

5 Spectral zeta, determinant coefficients, and cumulants

5.1 The exact spectral zeta function

Theorem 5.1 (Dyadic spectral zeta). *For $\Re s > 1$,*

$$\zeta_D(s) := \operatorname{Tr} D^{-s} = \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \quad (5.1)$$

This identity gives a meromorphic continuation to $\mathbb{C}$. Besides the pole at $s = 1$ with residue 2, it has the dyadic pole lattice

$$\omega_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z}, \quad (5.2)$$

with

$$\operatorname{Res}_{s=\omega_k} \zeta_D(s) = \frac{\zeta(\omega_k)}{\log 2}. \quad (5.3)$$

In particular,

$$\zeta_D(s) = -\frac{1}{2 \log 2} \frac{1}{s} - \frac{1}{4} - \frac{\log(2\pi)}{2 \log 2} + O(s) \quad (s \rightarrow 0). \quad (5.4)$$

Proof. The double sum in the proof of [theorem 3.1](#) gives [equation \(5.1\)](#). The pole at 1 is multiplied by $(1 - 2^{-1})^{-1} = 2$. At a zero ω_k of $1 - 2^{-s}$, its derivative is $(\log 2)2^{-s} = \log 2$, which gives [equation \(5.3\)](#). Expansion of $\zeta(s)$ and $(1 - e^{-s \log 2})^{-1}$ at 0 gives [equation \(5.4\)](#). $\square$

For $k \neq 0$, the numerator does not vanish on the imaginary axis; this follows, for example, from the functional equation together with the classical zero-free line $\Re s = 1$. The points ω_k therefore form a genuine vertical lattice of complex dimensions.

Corollary 5.2 (Reciprocal zero power sums). *For every integer $r \geq 1$,*

$$\sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^{2r}} = \frac{\zeta(2r)}{1 - 4^{-r}}. \quad (5.5)$$

5.2 Trace logarithm and a coefficient recurrence

For $|z| < 1$, the trace logarithm of the Fredholm determinant is absolutely convergent.

Theorem 5.3 (Spectral Taylor expansion). *One has*

$$\log \Phi(z) = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r(1-4^{-r})} z^{2r}, \quad |z| < 1. \quad (5.6)$$

If

$$\Phi(z) = \sum_{n=0}^{\infty} c_n z^{2n}, \quad c_0 = 1,$$

then

$$c_n = -\frac{1}{n} \sum_{r=1}^n \frac{\zeta(2r)}{1-4^{-r}} c_{n-r}. \quad (5.7)$$

In particular,

$$c_1 = -\frac{2\pi^2}{9}, \quad c_2 = \frac{38\pi^4}{2025}. \quad (5.8)$$

Proof. For $|z| < 1$,

$$\log \det(I - z^2 D^{-2}) = - \sum_{r \geq 1} \frac{z^{2r}}{r} \operatorname{Tr} D^{-2r},$$

and theorem 5.2 gives equation (5.6). Put $x = z^2$ and differentiate logarithmically:

$$x \frac{\sum_{n \geq 1} n c_n x^{n-1}}{\sum_{n \geq 0} c_n x^n} = - \sum_{r \geq 1} \frac{\zeta(2r)}{1-4^{-r}} x^r.$$

Coefficient comparison gives equation (5.7). □

5.3 A spectral interpretation of the known cumulants

Let X be the centered random variable in equation (2.2). Since $\mathbb{E}e^{itX} = \Phi(t/(2\pi))$, comparison of equation (5.6) with the cumulant expansion gives

$$\kappa_{2r}(X) = \frac{B_{2r}}{2r(1-4^{-r})}, \quad \kappa_{2r+1}(X) = 0. \quad (5.9)$$

These cumulants already occur in the frontier corpus. The new point is the identity

$$\kappa_{2r}(X) = (-1)^{r+1} \frac{(2r)!}{r(2\pi)^{2r}} \operatorname{Tr} D^{-2r}, \quad (5.10)$$

which interprets them as spectral power traces. For example $\kappa_2 = 1/9$ follows from $\operatorname{Tr} D^{-2} = 2\pi^2/9$.

6 Heat trace and dyadic complex dimensions

6.1 Mellin representation

Define

$$\Theta(t) = \operatorname{Tr} e^{-tD} = \sum_{m=1}^{\infty} a_m e^{-tm} = \sum_{j=0}^{\infty} \frac{1}{e^{2^j t} - 1}, \quad t > 0. \quad (6.1)$$

For $\Re s > 1$,

$$\int_0^\infty \Theta(t) t^{s-1} dt = \Gamma(s) \zeta_D(s) = \Gamma(s) \frac{\zeta(s)}{1-2^{-s}}. \quad (6.2)$$

The pole lattice equation (5.2) therefore produces logarithmic periodicity in physical scale.

Put

$$L = \log 2, \quad \omega_k = \frac{2\pi i k}{L},$$

and define the real-analytic, one-periodic function

$$\mathcal{P}_\Theta(u) = \frac{1}{L} \sum_{k \in \mathbb{Z} \setminus \{0\}} \Gamma(\omega_k) \zeta(\omega_k) e^{2\pi i k u}. \quad (6.3)$$

The exponential decay of $\Gamma(iy)$ makes the Fourier series absolutely and locally uniformly convergent with all derivatives.

6.2 Complete heat expansion

Theorem 6.1 (Heat trace with log-periodic oscillation). *Let*

$$C_\Theta = \frac{\gamma - \log(2\pi)}{2 \log 2} - \frac{1}{4}. \quad (6.4)$$

For every integer $J \geq 0$ and every $0 < \varepsilon < 1$,

$$\begin{aligned} \Theta(t) &= \frac{2}{t} - \frac{\log(1/t)}{2 \log 2} + C_\Theta + \mathcal{P}_\Theta\left(\log_2 \frac{1}{t}\right) \\ &\quad - \sum_{r=1}^J \frac{B_{2r}}{(2r)!(2^{2r-1} - 1)} t^{2r-1} + O_{J,\varepsilon}(t^{2J+1-\varepsilon}) \quad (t \downarrow 0). \end{aligned} \quad (6.5)$$

The first algebraic correction is $-t/12$.

Proof. Mellin inversion gives

$$\Theta(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \zeta_D(s) t^{-s} ds, \quad c > 1.$$

Move the contour left. The pole at $s = 1$ contributes $2/t$. At $s = 0$, $\Gamma(s) \zeta_D(s)$ has a double pole. From equation (5.4),

$$\Gamma(s) \zeta_D(s) = -\frac{1}{2L} \frac{1}{s^2} + \left(\frac{\gamma - \log(2\pi)}{2L} - \frac{1}{4} \right) \frac{1}{s} + O(1).$$

Multiplication by t^{-s} gives the logarithm and C_Θ . The residues at $s = \omega_k, k \neq 0$, sum to equation (6.3). At the negative odd integer $s = -(2r-1)$, the residue of Γ is $-1/(2r-1)!$ and $\zeta(1-2r) = -B_{2r}/(2r)$; this gives the displayed coefficient. The trivial zeros of ζ cancel the poles of Γ at negative even integers. Standard vertical-line bounds and the exponential decay of Γ justify the contour shift and remainder. $\square$

6.3 Size of the oscillation

The first Fourier coefficient in equation (6.3) is

$$\frac{\Gamma(2\pi i/L) \zeta(2\pi i/L)}{L} = -8.226408112794756 \times 10^{-7} - 9.760012263161017 \times 10^{-7} i, \quad (6.6)$$

with modulus $1.276446747088608 \times 10^{-6}$. Thus the oscillation is small but exact. It is not an error term: it is a Fourier series forced by the pole lattice of ζ_D .

Remark 6.2 (Two kinds of self-similar fluctuation). The exact counting remainder $-s_2(M)$ is discontinuous and arithmetic. The heat remainder $\mathcal{P}_\Theta(\log_2(1/t))$ is smooth and analytic. They are two transforms of the same dyadic spectral multiplicity: a sharp cutoff retains binary carries, whereas exponential smoothing turns the scale lattice into an analytic periodic wave.

7 Exact local jets at all integer zeros

7.1 A normalized factorization

Fix a nonzero integer m . Write

$$r = \nu_2(|m|), \quad a = r + 1, \quad q = \frac{m}{2^r}, \quad (7.1)$$

so that q is odd.

Theorem 7.1 (Exact all-order zero jet). *For w in a neighborhood of 0,*

$$\Phi(m + w) = C_m w^a R_m(w), \quad (7.2)$$

where

$$C_m = -\frac{\Phi(q/2)}{m^a} \quad (7.3)$$

and

$$R_m(w) = \left(\frac{m}{m+w}\right)^a \prod_{h=0}^r \operatorname{sinc}\left(\frac{\pi w}{2^h}\right) \frac{\Phi(q/2 + w/2^{r+1})}{\Phi(q/2)}. \quad (7.4)$$

In particular R_m is analytic and $R_m(0) = 1$, and every derivative is given exactly by

$$\Phi^{(a+\ell)}(m) = C_m (a+\ell)! [w^\ell] R_m(w), \quad \ell \geq 0. \quad (7.5)$$

The first nonzero derivative is

$$\Phi^{(a)}(m) = -\frac{a!}{m^a} \Phi(q/2). \quad (7.6)$$

Proof. For $0 \leq h \leq r$, $m/2^h$ is an integer and

$$\operatorname{sinc}\left(\frac{\pi(m+w)}{2^h}\right) = (-1)^{m/2^h} \frac{w}{m+w} \operatorname{sinc}\left(\frac{\pi w}{2^h}\right).$$

The product of the signs is

$$(-1)^{\sum_{h=0}^r m/2^h} = (-1)^{q(2^{r+1}-1)} = -1.$$

The tail of the original sinc product is

$$\prod_{h=r+1}^{\infty} \operatorname{sinc}\left(\frac{\pi(m+w)}{2^h}\right) = \Phi\left(\frac{q}{2} + \frac{w}{2^{r+1}}\right).$$

Multiplying these identities gives

$$\Phi(m+w) = -\left(\frac{w}{m+w}\right)^a \prod_{h=0}^r \operatorname{sinc}\left(\frac{\pi w}{2^h}\right) \Phi\left(\frac{q}{2} + \frac{w}{2^{r+1}}\right),$$

which is equivalent to equation (7.2)–equation (7.4). Since $q/2$ is not an integer, $\Phi(q/2) \neq 0$. Coefficient extraction proves equation (7.5). $\square$

7.2 The next coefficient and dyadic-ray recursion

Differentiating equation (7.4) at 0 gives

$$R'_m(0) = -\frac{a}{m} + \frac{1}{2^{r+1}} \frac{\Phi'(q/2)}{\Phi(q/2)}. \quad (7.7)$$

Therefore

$$\Phi^{(a+1)}(m) = -\frac{(a+1)!\Phi(q/2)}{m^a} \left(-\frac{a}{m} + \frac{1}{2^{r+1}} \frac{\Phi'(q/2)}{\Phi(q/2)} \right). \quad (7.8)$$

All higher jets follow by formal multiplication of the three analytic factors in equation (7.4).

The leading coefficients propagate exactly along dyadic rays.

Corollary 7.2 (Dyadic-ray recursion). *If $a = 1 + \nu_2(m)$, then*

$$C_{2m} = \frac{C_m}{2^{a+1}m}. \quad (7.9)$$

Equivalently,

$$\frac{\Phi^{(a+1)}(2m)}{(a+1)!} = \frac{1}{2^{a+1}m} \frac{\Phi^{(a)}(m)}{a!}. \quad (7.10)$$

Proof. The integers m and $2m$ have the same odd part, while their multiplicities differ by one. Substitute into equation (7.3). $\square$

7.3 Thue–Morse signs of the leading jets

For $m > 0$, write its odd part as $q = 2k+1$. The established interval-sign theorem gives $\text{sgn } \Phi(q/2) = (-1)^{s_2(k)}$. Hence

$$\boxed{\text{sgn } \Phi^{(1+\nu_2(m))}(m) = -(-1)^{s_2((q-1)/2)}}. \quad (7.11)$$

Thus the leading Taylor coefficient at every positive zero is controlled by the Thue–Morse bit of half the predecessor of its odd part.

7.4 Numerical checks

The following values were computed independently from a high-precision truncated sinc product and from equation (7.6); the two evaluations agreed to at least 70 decimal digits before rounding.

m	$a = 1 + \nu_2(m)$	$\Phi^{(a)}(m)$	sign
1	1	−0.5537712758888100953	−
2	2	−0.2768856379444050477	−
3	1	0.01540076637810223766	+
4	3	−0.05191605711457594644	−
6	2	0.002566794396350372944	+
8	4	−0.003244753569660996652	−
12	3	0.0001604246497718983090	+

These computations are checks, not substitutes for the proof.

8 The dyadic q -Euler determinant

8.1 Definition and q -Pochhammer tower

Fix $0 < Q < 1$. Since Q^D is trace class, define

$$\mathcal{E}(u; Q) = \det(I - uQ^D). \quad (8.1)$$

Theorem 8.1 (Dyadic q -Euler product). *For every $u \in \mathbb{C}$,*

$$\mathcal{E}(u; Q) = \prod_{m=1}^{\infty} (1 - uQ^m)^{a_m} \quad (8.2)$$

$$= \prod_{j=0}^{\infty} (uQ^{2^j}; Q^{2^j})_{\infty}. \quad (8.3)$$

It satisfies the exact renormalization equation

$$\boxed{\mathcal{E}(u; Q) = (uQ; Q)_{\infty} \mathcal{E}(u; Q^2).} \quad (8.4)$$

Proof. The first product is the eigenvalue formula for a Fredholm determinant. The decomposition $D \simeq N \oplus 2D$ gives

$$\det(I - uQ^D) = \det(I - uQ^N) \det(I - uQ^{2D}) = (uQ; Q)_{\infty} \mathcal{E}(u; Q^2).$$

Iteration gives equation (8.3); convergence follows from $\sum_{j,n} |u|Q^{2^j n} < \infty$. $\square$

8.2 Lambert series and plethystic logarithm

For $|u|Q < 1$,

$$-\log \mathcal{E}(u; Q) = \sum_{\ell=1}^{\infty} \frac{u^{\ell}}{\ell} \sum_{j=0}^{\infty} \frac{Q^{2^j \ell}}{1 - Q^{2^j \ell}}. \quad (8.5)$$

Equivalently, the ordinary generating function of the zero multiplicities is the dyadic Lambert series

$$\boxed{\sum_{m=1}^{\infty} a_m x^m = \sum_{j=0}^{\infty} \frac{x^{2^j}}{1 - x^{2^j}}, \quad |x| < 1.} \quad (8.6)$$

The same self-similarity now appears in three transforms:

$$a_m \quad \longleftrightarrow \quad \frac{\zeta(s)}{1 - 2^{-s}} \quad \longleftrightarrow \quad \sum_{j \geq 0} \frac{x^{2^j}}{1 - x^{2^j}}.$$

9 Finite multiscale Gaussian-binomial formulae

9.1 A finite determinant

For $M \geq 1$, define

$$\mathcal{E}_M(u; Q) = \prod_{m=1}^M (1 - uQ^m)^{a_m}. \quad (9.1)$$

Put

$$J = \lfloor \log_2 M \rfloor, \quad L_j = \left\lfloor \frac{M}{2^j} \right\rfloor.$$

Theorem 9.1 (Finite multiscale factorization). *One has*

$$\mathcal{E}_M(u; Q) = \prod_{j=0}^J (uQ^{2^j}; Q^{2^j})_{L_j}. \quad (9.2)$$

Its degree in u is

$$\deg_u \mathcal{E}_M = A(M) = 2M - s_2(M). \quad (9.3)$$

Writing

$$\mathcal{E}_M(u; Q) = \sum_{k=0}^{A(M)} (-1)^k C_{M,k}(Q) u^k, \quad (9.4)$$

one has the explicit Gaussian-binomial convolution

$$C_{M,k}(Q) = \sum_{\substack{k_0 + \dots + k_J = k \\ 0 \leq k_j \leq L_j}} Q^{\sum_{j=0}^J 2^j \binom{k_j+1}{2}} \prod_{j=0}^J \begin{bmatrix} L_j \\ k_j \end{bmatrix}_{Q^{2^j}}. \quad (9.5)$$

Proof. Expand a_m as the number of representations $m = 2^j n$ and regroup:

$$\mathcal{E}_M(u; Q) = \prod_{j=0}^J \prod_{n=1}^{L_j} (1 - uQ^{2^j n}),$$

which is equation (9.2). Its degree is $\sum_j L_j = A(M)$. Applying equation (2.4) to each factor gives

$$(uQ^{2^j}; Q^{2^j})_{L_j} = \sum_{k_j=0}^{L_j} (-1)^{k_j} Q^{2^j \binom{k_j+1}{2}} \begin{bmatrix} L_j \\ k_j \end{bmatrix}_{Q^{2^j}} u^{k_j}.$$

Multiplication and collection of the total degree proves equation (9.5). $\square$

9.2 Dyadic endpoints and a multiscale factorial

For $M = 2^n - 1$,

$$L_j = 2^{n-j} - 1, \quad A(2^n - 1) = 2^{n+1} - n - 2. \quad (9.6)$$

The top Q -degree of equation (9.1) is

$$B(M) := \sum_{m=1}^M m a_m = \frac{1}{2} \sum_{j=0}^J 2^j L_j (L_j + 1). \quad (9.7)$$

At a dyadic endpoint,

$$B(2^n - 1) = 4^n - (n + 2)2^{n-1}. \quad (9.8)$$

The classical limit of the normalizing product is

$$\lim_{Q \uparrow 1} \frac{\mathcal{E}_M(1; Q)}{(1 - Q)^{A(M)}} = \prod_{m=1}^M m^{a_m} = \prod_{j=0}^J 2^{jL_j} L_j!. \quad (9.9)$$

Thus the finite determinant naturally defines a dyadic multiscale factorial.

10 Mellin asymptotics of the q -Pochhammer tower

Put $Q = e^{-t}$, $t > 0$, and

$$y = \log \frac{1}{t}, \quad u_t = \log_2 \frac{1}{t} = \frac{y}{\log 2}.$$

The identity equation (8.5) may be rewritten as

$$-\log \mathcal{E}(u; e^{-t}) = \sum_{\ell=1}^{\infty} \frac{u^{\ell}}{\ell} \Theta(\ell t). \quad (10.1)$$

This transfers the heat complex dimensions directly to a tower of q -Pochhammer symbols.

10.1 The subcritical regime $|u| < 1$

Let

$$\mathcal{P}_u(v) = \frac{1}{\log 2} \sum_{k \neq 0} \Gamma(\omega_k) \zeta(\omega_k) \text{Li}_{1+\omega_k}(u) e^{2\pi i k v}. \quad (10.2)$$

For u in a compact subset of the unit disk this series is absolutely convergent and analytic in both u and v .

Theorem 10.1 (Subcritical q -Euler asymptotic). *Fix $\rho < 1$. Uniformly for $|u| \leq \rho$, every $J \geq 0$, and every $0 < \varepsilon < 1$,*

$$\begin{aligned} -\log \mathcal{E}(u; e^{-t}) &= \frac{2 \text{Li}_2(u)}{t} + \frac{\log(1-u)}{2 \log 2} y + C(u) + \mathcal{P}_u(u_t) \\ &\quad - \sum_{r=1}^J \frac{B_{2r} \text{Li}_{2-2r}(u)}{(2r)!(2^{2r-1} - 1)} t^{2r-1} + O_{J,\rho,\varepsilon}(t^{2J+1-\varepsilon}), \end{aligned} \quad (10.3)$$

where

$$C(u) = -\frac{1}{2 \log 2} \left. \frac{\partial}{\partial s} \text{Li}_s(u) \right|_{s=1} - C_{\Theta} \log(1-u). \quad (10.4)$$

Proof. For $c > 1$, Mellin inversion and absolute convergence give

$$-\log \mathcal{E}(u; e^{-t}) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \text{Li}_{s+1}(u) \zeta_D(s) t^{-s} ds. \quad (10.5)$$

At $s = 1$ the residue is $2 \text{Li}_2(u)/t$. At $s = 0$, write

$$\zeta_D(s) = \frac{a_{-1}}{s} + a_0 + O(s), \quad a_{-1} = -\frac{1}{2 \log 2}, \quad a_0 = -\frac{1}{4} - \frac{\log(2\pi)}{2 \log 2}.$$

The double-pole residue after multiplication by t^{-s} is

$$\frac{\log(1-u)}{2 \log 2} y - \frac{1}{2 \log 2} \left. \frac{\partial}{\partial s} \text{Li}_s(u) \right|_{s=1} - C_{\Theta} \log(1-u).$$

The poles ω_k give equation (10.2). At $s = -(2r-1)$, the residue equals the displayed Bernoulli-polylogarithm coefficient. At negative even integers the trivial zeros of ζ cancel the poles of Γ . $\square$

The expansion is nonuniform at $u = 1$: $\text{Li}_1(u) = -\log(1-u)$ diverges, and a second singular factor meets the pole at $s = 0$. The critical limit must be calculated separately.

10.2 The critical product $u = 1$

Define

$$\mathcal{P}_E(v) = \frac{1}{\log 2} \sum_{k \neq 0} \Gamma(\omega_k) \zeta(1 + \omega_k) \zeta(\omega_k) e^{2\pi i k v}. \quad (10.6)$$

Let γ_1 be the first Stieltjes constant in

$$\zeta(1 + s) = \frac{1}{s} + \gamma - \gamma_1 s + O(s^2).$$

Put

$$C_E = \frac{\zeta''(0) + \gamma_1 + \gamma^2/2 - \pi^2/12}{2 \log 2} - \frac{\log(2\pi)}{4} - \frac{\log 2}{24}. \quad (10.7)$$

Numerically, $C_E = -2.4612722039168654768 \dots$

Theorem 10.2 (Critical dyadic q -Pochhammer tower). *For every $N > 0$,*

$$\begin{aligned} -\log \mathcal{E}(1; e^{-t}) &= \frac{\pi^2}{3t} - \frac{y^2}{4 \log 2} - \left(\frac{1}{4} + \frac{\log(2\pi)}{2 \log 2} \right) y \\ &\quad + C_E + \mathcal{P}_E(u_t) + \frac{t}{24} + O_N(t^N) \quad (t \downarrow 0). \end{aligned} \quad (10.8)$$

Equivalently, this is the asymptotic of

$$\mathcal{E}(1; e^{-t}) = \prod_{j=0}^{\infty} (e^{-2^j t}; e^{-2^j t})_{\infty}. \quad (10.9)$$

Proof. At $u = 1$, equation (10.5) becomes

$$\frac{1}{2\pi i} \int \Gamma(s) \zeta(s+1) \frac{\zeta(s)}{1-2^{-s}} t^{-s} ds.$$

The pole at $s = 1$ contributes $2\zeta(2)/t = \pi^2/(3t)$. At $s = 0$ the three factors $\Gamma(s)$, $\zeta(s+1)$, and $(1-2^{-s})^{-1}$ produce a triple pole. Write

$$\Gamma(s) \zeta(1+s) = \frac{1}{s^2} + H_0 + O(s), \quad H_0 = -\gamma_1 - \frac{\gamma^2}{2} + \frac{\pi^2}{12},$$

and

$$\zeta_D(s) = \frac{a_{-1}}{s} + a_0 + a_1 s + O(s^2), \quad a_1 = \frac{\zeta''(0)}{2 \log 2} - \frac{\log(2\pi)}{4} - \frac{\log 2}{24}.$$

The coefficient of $1/s$ after multiplication by e^{sy} is

$$\frac{a_{-1}}{2} y^2 + a_0 y + a_1 + a_{-1} H_0,$$

which is exactly the quadratic-log part and C_E . The poles ω_k give equation (10.6). The pole at $s = -1$ contributes $t/24$. Every pole of Γ at an integer $s = -n$, $n \geq 2$, is canceled by a trivial zero of either $\zeta(s)$ or $\zeta(s+1)$. The contour can therefore be moved arbitrarily far left, giving $O_N(t^N)$. $\square$

Remark 10.3. The quadratic logarithm is a critical phenomenon. For $|u| < 1$ the origin is a double Mellin pole and produces a term linear in y ; at $u = 1$, the pole of $\zeta(s+1)$ raises the order to three and produces $-y^2/(4 \log 2)$.

11 A symmetric q -integer deformation of the sinc product

11.1 Symmetric q -integers

For $0 < Q < 1$, define

$$[m]_Q = \frac{Q^{-m/2} - Q^{m/2}}{Q^{-1/2} - Q^{1/2}} = Q^{-(m-1)/2} \frac{1 - Q^m}{1 - Q}. \quad (11.1)$$

If $Q = e^{-t}$, then

$$[m]_Q = \frac{\sinh(mt/2)}{\sinh(t/2)}. \quad (11.2)$$

Thus $[m]_Q \geq m$, $[m]_Q \rightarrow m$ as $Q \uparrow 1$, and $[m]_Q$ grows exponentially in m for fixed $Q < 1$. The multiplicative identity

$$[ab]_Q = [a]_Q [b]_{Q^a} \quad (11.3)$$

will encode exact q -self-similarity.

By functional calculus, let $[D]_Q$ be diagonal with eigenvalues $[2^j n]_Q$.

Definition 11.1 (Symmetric dyadic q -sinc determinant). Define

$$\Phi_Q(z) = \det(I - z^2 [D]_Q^{-2}) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{[m]_Q^2} \right)^{a_m}. \quad (11.4)$$

Also define the one-scale factor

$$S_Q(z) = \det(I - z^2 [N]_Q^{-2}) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{[n]_Q^2} \right). \quad (11.5)$$

Both are entire functions of order zero for fixed $Q < 1$.

11.2 Exact q -scaling

Theorem 11.2 (q -renormalization equation). *The functional calculus version of $D \simeq N \oplus 2D$ is*

$$[D]_Q \simeq [N]_Q \oplus [2]_Q [D]_{Q^2}. \quad (11.6)$$

Consequently,

$$\Phi_Q(z) = S_Q(z) \Phi_{Q^2} \left(\frac{z}{[2]_Q} \right). \quad (11.7)$$

Iterating,

$$\Phi_Q(z) = \prod_{j=0}^{\infty} S_{Q^{2^j}} \left(\frac{z}{[2^j]_Q} \right). \quad (11.8)$$

As $Q \uparrow 1$, this becomes the classical identity

$$\Phi(z) = \prod_{j=0}^{\infty} \text{sinc}(\pi z / 2^j).$$

Proof. On the $j \geq 1$ subspace, the eigenvalues are $[2(2^{j-1}n)]_Q$. By equation (11.3),

$$[2(2^{j-1}n)]_Q = [2]_Q [2^{j-1}n]_{Q^2}.$$

This proves equation (11.6). Apply determinant multiplicativity to get equation (11.7); repeated use and

$$[2^j]_Q = \prod_{h=0}^{j-1} [2]_{Q^{2^h}}$$

give equation (11.8). □

11.3 Exact q -Pochhammer factorization

Set

$$\eta = \frac{z(1-Q)}{\sqrt{Q}}, \quad \lambda + \lambda^{-1} = 2 + \eta^2. \quad (11.9)$$

Either root may be chosen; every formula below is symmetric under $\lambda \leftrightarrow \lambda^{-1}$.

Theorem 11.3 (q -Pochhammer representation of Φ_Q). *For every $m \geq 1$,*

$$1 - \frac{z^2}{[m]_Q^2} = \frac{(1 - \lambda Q^m)(1 - \lambda^{-1} Q^m)}{(1 - Q^m)^2}. \quad (11.10)$$

Therefore

$$\Phi_Q(z) = \frac{\mathcal{E}(\lambda; Q) \mathcal{E}(\lambda^{-1}; Q)}{\mathcal{E}(1; Q)^2}. \quad (11.11)$$

Equivalently,

$$\Phi_Q(z) = \prod_{j=0}^{\infty} \frac{(\lambda Q^{2^j}; Q^{2^j})_{\infty} (\lambda^{-1} Q^{2^j}; Q^{2^j})_{\infty}}{(Q^{2^j}; Q^{2^j})_{\infty}^2}. \quad (11.12)$$

Likewise,

$$S_Q(z) = \frac{(\lambda Q; Q)_{\infty} (\lambda^{-1} Q; Q)_{\infty}}{(Q; Q)_{\infty}^2}. \quad (11.13)$$

Proof. From equation (11.1),

$$\frac{z^2}{[m]_Q^2} = \frac{\eta^2 Q^m}{(1 - Q^m)^2}.$$

The numerator after subtraction is

$$(1 - Q^m)^2 - \eta^2 Q^m = 1 - (2 + \eta^2) Q^m + Q^{2m} = (1 - \lambda Q^m)(1 - \lambda^{-1} Q^m),$$

which proves equation (11.10). Raise to the multiplicity a_m , multiply over m , and use equation (8.2) and equation (8.3). □

Direct bridge to the classical sinc product. Combining theorem 11.3 with the convergence theorem below gives the explicit limit

$$\Phi(z) = \lim_{Q \uparrow 1} \prod_{j=0}^{\infty} \frac{(\lambda_Q(z) Q^{2^j}; Q^{2^j})_{\infty} (\lambda_Q(z)^{-1} Q^{2^j}; Q^{2^j})_{\infty}}{(Q^{2^j}; Q^{2^j})_{\infty}^2},$$

where $\lambda_Q(z) + \lambda_Q(z)^{-1} = 2 + z^2(1 - Q)^2/Q$. This is a direct q -Pochhammer representation of the infinite sinc product through a singular but controlled $Q \uparrow 1$ limit.

11.4 Convergence to the Rvachev Fourier transform

Theorem 11.4 (Trace-norm and locally uniform limit). *As $Q \uparrow 1$,*

$$\|[D]_Q^{-2} - D^{-2}\|_1 \longrightarrow 0. \quad (11.14)$$

Consequently,

$$\boxed{\Phi_Q(z) \longrightarrow \Phi(z)} \quad (11.15)$$

locally uniformly on $\mathbb{C}$.

Proof. The eigenvalues satisfy $[m]_Q \geq m$ and converge pointwise to m . Hence

$$\|[D]_Q^{-2} - D^{-2}\|_1 = \sum_{m=1}^{\infty} a_m \left(\frac{1}{m^2} - \frac{1}{[m]_Q^2} \right) \longrightarrow 0$$

by dominated convergence, dominated by a_m/m^2 . Continuity of the Fredholm determinant with respect to trace norm, uniformly when z ranges over compact sets, proves the result. $\square$

11.5 Exact q -zero arithmetic and a Thue–Morse sign law

Theorem 11.5 (q -zero spectrum). *For fixed $0 < Q < 1$, the nonzero zeros of Φ_Q are exactly*

$$z = \pm [m]_Q, \quad m \geq 1, \quad (11.16)$$

with multiplicity $a_m = 1 + \nu_2(m)$. Hence

$$\boxed{\operatorname{sgn} \Phi_Q(x) = (-1)^{s_2(M)} \quad \text{for } [M]_Q < x < [M+1]_Q.} \quad (11.17)$$

If $Q = e^{-t}$, define

$$M_Q(R) = \left\lfloor \frac{2}{t} \operatorname{arsinh} \left(R \sinh \frac{t}{2} \right) \right\rfloor. \quad (11.18)$$

Then the number of nonzero zeros in $|z| \leq R$, counted with multiplicity, is

$$\boxed{\mathcal{N}_{\Phi_Q}(R) = 4M_Q(R) - 2s_2(M_Q(R)).} \quad (11.19)$$

For fixed $Q < 1$, $\mathcal{N}_{\Phi_Q}(R) = O(\log R)$ and Φ_Q has order zero. The limit $Q \uparrow 1$ is a singular transition to the linear zero growth and order one of Φ .

Proof. The sequence $[m]_Q$ is strictly increasing. The product equation (11.4) gives the zero set and multiplicities. Crossing the first M positive zeros changes sign by $(-1)^{A(M)} = (-1)^{s_2(M)}$. Formula equation (11.2) shows that $[m]_Q \leq R$ exactly when

$$m \leq \frac{2}{t} \operatorname{arsinh}(R \sinh(t/2)),$$

which proves the count. Exponential growth of $[m]_Q$ gives logarithmic counting and order zero. $\square$

11.6 The q -spectral zeta and coefficient recurrence

For $\Re s > 0$, define

$$\zeta_{D,Q}(s) = \operatorname{Tr}[D]_Q^{-s} = \sum_{m=1}^{\infty} \frac{a_m}{[m]_Q^s}. \quad (11.20)$$

Let

$$\mathcal{A}(x) = \sum_{m=1}^{\infty} a_m x^m = \sum_{j=0}^{\infty} \frac{x^{2^j}}{1 - x^{2^j}}. \quad (11.21)$$

Proposition 11.6 (q -spectral Lambert expansion). *For $\Re s > 0$,*

$$\zeta_{D,Q}(s) = (1-Q)^s Q^{-s/2} \sum_{k=0}^{\infty} \frac{(s)_k}{k!} \mathcal{A}(Q^{k+s/2}), \quad (11.22)$$

where $(s)_k = s(s+1) \cdots (s+k-1)$. For $\Re s > 1$,

$$\zeta_{D,Q}(s) \longrightarrow \zeta_D(s) \quad (Q \uparrow 1). \quad (11.23)$$

If

$$\Phi_Q(z) = \sum_{n=0}^{\infty} c_n(Q) z^{2n},$$

then

$$c_0(Q) = 1, \quad c_n(Q) = -\frac{1}{n} \sum_{r=1}^n \zeta_{D,Q}(2r) c_{n-r}(Q). \quad (11.24)$$

Proof. From equation (11.1),

$$[m]_Q^{-s} = (1-Q)^s Q^{-s/2} Q^{ms/2} (1-Q^m)^{-s}.$$

Expand $(1-Q^m)^{-s} = \sum_{k \geq 0} (s)_k Q^{mk} / k!$ and sum over m . Dominated convergence using $[m]_Q \geq m$ proves equation (11.23). Finally,

$$\log \Phi_Q(z) = - \sum_{r \geq 1} \frac{\zeta_{D,Q}(2r)}{r} z^{2r} \quad (|z| < 1),$$

and logarithmic differentiation gives equation (11.24). $\square$

12 A base- b extension

The operator mechanism is not restricted to 2. Let $b \geq 2$ be an integer and define

$$D_b e_{j,n} = b^j n e_{j,n}, \quad \Phi_b(z) = \det(I - z^2 D_b^{-2}) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{b^j}\right). \quad (12.1)$$

Let $\nu_b(m) = \max\{j : b^j \mid m\}$ and $s_b(M)$ be the sum of the base- b digits of M .

Theorem 12.1 (Base- b spectral package). *One has*

$$D_b \simeq N \oplus b D_b, \quad (12.2)$$

$$\operatorname{mult}_{D_b}(m) = 1 + \nu_b(m), \quad (12.3)$$

$$\zeta_{D_b}(s) = \frac{\zeta(s)}{1 - b^{-s}}, \quad (12.4)$$

$$A_b(M) := \sum_{m \leq M} (1 + \nu_b(m)) = \sum_{j \geq 0} \left\lfloor \frac{M}{b^j} \right\rfloor = \frac{bM - s_b(M)}{b-1}. \quad (12.5)$$

For symmetric q -integers,

$$\Phi_{b,Q}(z) = \prod_{m \geq 1} \left(1 - \frac{z^2}{[m]_Q^2}\right)^{1 + \nu_b(m)} \quad (12.6)$$

satisfies

$$\Phi_{b,Q}(z) = S_Q(z) \Phi_{b,Q^b}(z/[b]_Q). \quad (12.7)$$

The associated q -Euler determinant is

$$\mathcal{E}_b(u; Q) = \prod_{j=0}^{\infty} (uQ^{b^j}; Q^{b^j})_{\infty}. \quad (12.8)$$

Proof. All statements follow by splitting off the $j = 0$ summand and relabeling $j \mapsto j - 1$. The digit formula uses

$$\sum_{j \geq 1} \left\lfloor \frac{M}{b^j} \right\rfloor = \frac{M - s_b(M)}{b - 1}.$$

The q -scaling follows from $[bn]_Q = [b]_Q [n]_{Q^b}$. □

The dyadic case is distinguished because the parity of [equation \(12.5\)](#) collapses to

$$(-1)^{A_2(M)} = (-1)^{s_2(M)},$$

which is exactly Thue–Morse. Thus the Thue–Morse sign is not an accidental decoration: it is the parity character singled out by the base-2 spectral tower.

13 Computational verification and reproducible checks

13.1 What was checked

The following identities were independently evaluated at high precision:

1. the leading zero-jet formula [equation \(7.6\)](#) for integers of several distinct 2-adic valuations;
2. the factorization [equation \(11.11\)](#) by comparing the direct zero product and the quotient of dyadic q -Euler determinants;
3. convergence $\Phi_Q(z) \rightarrow \Phi(z)$ at real points inside and outside the first zero interval;
4. the heat expansion [equation \(6.5\)](#), including the coefficient $-1/12$ of t and the first Fourier coefficient of $\mathcal{P}_{\Theta}$;
5. the critical expansion [equation \(10.8\)](#), whose residual after the displayed $t/24$ term was below 6×10^{-12} in the tested range $0.01 \leq t \leq 0.2$ with a finite Fourier truncation.

13.2 $q \uparrow 1$ sample values

Q	$\Phi_Q(0.3)$	$\Phi_Q(1.2)$	$\Phi_Q(2.7)$
0.80	0.8316534376	−0.08646646	−0.03017980
0.90	0.8241751473	−0.07432843	−0.01331576
0.95	0.8204894677	−0.06907163	−0.00902981
0.98	0.8182876536	−0.06613499	−0.00721005
0.99	0.8175541599	−0.06518866	−0.00669771
$Q = 1$ limit	0.8168198779	−0.0642567960	−0.0062253251

The convergence is not claimed to be monotone in general; the table merely illustrates the trace-norm theorem.

13.3 A practical evaluation hierarchy

The new identities suggest several complementary algorithms:

- (a) **Near the origin:** use the coefficient recurrence [equation \(5.7\)](#) or its q -version [equation \(11.24\)](#).
- (b) **Near an integer zero:** use the normalized local factor [equation \(7.2\)](#); this removes catastrophic cancellation and exposes the exact multiplicity.
- (c) **On compact sets away from zeros:** use the finite canonical product with the certified tail estimate [equation \(4.9\)](#).
- (d) **For $Q < 1$:** use the q -Pochhammer quotient [equation \(11.12\)](#), exploiting fast exponential convergence at large scales.
- (e) **For Q close to 1:** combine the exact quotient with the critical asymptotic [equation \(10.8\)](#) to cancel large denominator terms analytically before numerical evaluation.

14 Connections back to the Fabius and Thue–Morse theories

14.1 Two independent appearances of q -binomial structure

The existing Fabius corpus derives base-1/2 Gaussian-binomial coefficients from finite dyadic coefficient extraction and exact values of F at dyadic rationals. The present report derives Gaussian-binomial coefficients from a different object: finite spectral determinants of the zero divisor of $\widehat{\text{up}}$. The common mechanism is multiscale dyadic refinement:

$$\text{dyadic value formulas} \longleftarrow \text{refinement blocks} \longrightarrow \text{spectral multiplicity blocks}.$$

The two coefficient systems are not identical, but they live in the same category of products indexed by powers of 2. This opens a concrete comparison problem: identify transforms that carry the exact-value q -binomial arrays into the spectral arrays $C_{M,k}(Q)$ of [equation \(9.5\)](#).

14.2 Thue–Morse as a determinant sign

Equation [equation \(4.5\)](#) can be rewritten

$$\tau_M = \text{sgn det}(I - x^2 D^{-2}) \quad (M < x < M + 1). \quad (14.1)$$

Thus a binary automatic sequence is recovered from signs of a single entire Fredholm determinant. The q -deformed version [equation \(11.17\)](#) moves the zero locations but preserves the automatic sign word exactly.

14.3 Moments and zeros from one spectrum

The same spectrum produces both:

- the zero multiplicities through its eigenvalue degeneracies;
- the probability cumulants through the traces $\text{Tr } D^{-2r}$.

This is stronger than a formal analogy. It says the local probability law of the Rvachev density and the global zero divisor of its Fourier transform are two evaluations of the same spectral data.

14.4 Heat smoothing versus Fourier decay

The active Fourier drafts study $\Phi(x)$ at large real x , where lacunary products create pointwise exceptional behavior and metric limit laws. The heat trace $\Theta(t)$ studies the same multiplicities under exponential smoothing. It is much more rigid: all oscillations are captured by the explicit vertical lattice $2\pi i\mathbb{Z}/\log 2$. A promising next step is to seek a transform connecting the log-periodic heat term to the periodic amplitudes that occur in endpoint asymptotics of F .

15 Lean formalization roadmap

The operator-theoretic presentation is conceptually compact, but a successful Lean formalization should proceed from finite identities to analytic limits.

15.1 Layer 1: arithmetic and finite products

The most accessible statements are:

1. $a_m = 1 + \nu_2(m)$ and the recursions $a_{2m} = a_m + 1$, $a_{2m+1} = 1$;
2. $\sum_{m \leq M} a_m = 2M - s_2(M)$;
3. the finite factorization [equation \(9.2\)](#);
4. the Gaussian-binomial coefficient formula [equation \(9.5\)](#);
5. the finite form of the single-factor identity [equation \(11.10\)](#);
6. the base- b digit identity [equation \(12.5\)](#).

These require no infinite-dimensional analysis.

15.2 Layer 2: locally finite zero factorization

The exact local-jet theorem can be formalized directly from the sinc product by splitting it at $r = \nu_2(m)$. The crucial finite identity is

$$\prod_{h=0}^r (-1)^{m/2^h} = -1.$$

The analytic tail is just a rescaled copy of Φ . This route avoids a general theory of Fredholm determinants.

15.3 Layer 3: infinite products and traces

One can next formalize:

1. absolute convergence of $\sum_{j,n} (2^j n)^{-2}$;
2. the regrouped canonical product;
3. the Dirichlet-series identity $\zeta_D(s) = \zeta(s)/(1 - 2^{-s})$ on $\Re s > 1$;
4. trace-norm convergence of the q -deformation, represented as convergence of summable diagonal sequences.

Only after these are stable is it efficient to introduce Mathlib's operator determinant infrastructure, if the available API is sufficiently mature.

15.4 Layer 4: Mellin asymptotics

The heat and q -Euler expansions require contour shifting, residues, zeta continuation, and vertical estimates. They are the deepest formalization target. A useful intermediate milestone is a theorem schema: given a meromorphic Mellin transform with a finite specified pole set and suitable vertical decay, derive an asymptotic expansion. The dyadic results would then become applications.

16 Frontier questions

Problem 16.1 (Arithmetic of half-integer amplitudes). Classify or sharply estimate the numbers $\Phi((2k+1)/2)$ that govern every leading integer-zero jet through [equation \(7.3\)](#). Determine whether their signs and 2-adic or transcendence properties admit formulae parallel to the exact dyadic values of F .

Problem 16.2 (Comparison of the two Gaussian-binomial webs). Construct an explicit transform between the coefficient arrays in the exact dyadic-value formulas for F and the spectral coefficients $C_{M,k}(Q)$ in [equation \(9.5\)](#). The shared multiscale factorization suggests a common incidence algebra or a common family of refinement matrices.

Problem 16.3 (Critical double scaling). Analyze $\Phi_Q(z)$ when $Q = e^{-t} \uparrow 1$ and z grows with t^{-1} . The fixed- Q function has order zero and logarithmic zero growth, while the limit has order one and linear zero growth. Identify the transition kernels and scaling regimes.

Problem 16.4 (Heat periodicity and endpoint periodicity). The heat trace has the explicit periodic wave [equation \(6.3\)](#); endpoint asymptotics of the Fabius function also contain periodic dependence on a dyadic logarithmic phase. Determine whether both are projections of a single meromorphic two-variable transform.

Problem 16.5 (Twisted spectral traces). Let $Je_{j,n} = \tau_n e_{j,n}$. For $\Re s > 1$,

$$\mathrm{Tr}(JD^{-s}) = \frac{1}{1-2^{-s}} \sum_{n=1}^{\infty} \frac{\tau_n}{n^s}. \quad (16.1)$$

Study its continuation, functional relations, and whether its singularities encode known Thue–Morse Dirichlet-series phenomena in a spectral form.

Problem 16.6 (Transfer operators). Relate the diagonal spectral model D to the transfer operators used in the Fourier-decay corpus. The former captures the exact zero divisor and smoothed scale statistics; the latter captures real-axis orbit statistics. A common operator category could unify zeros, decay, and probabilistic limit laws.

Conjecture 16.7 (Universality of the symmetric q -deformation). *Among deformations that (i) preserve the multiplicity word $1 + \nu_2(m)$, (ii) obey an exact two-scale factorization, (iii) have order zero for fixed $Q < 1$, and (iv) converge locally uniformly to Φ , the symmetric q -integer model [equation \(11.4\)](#) is unique up to an analytic reparameterization of Q and a scalar rescaling of z .*

17 Conclusion

The infinite sinc product of Rvachev’s up-function admits a spectral organization that is invisible when the product is treated only factor by factor. The diagonal operator D turns dyadic refinement into the unitary decomposition $D \simeq N \oplus 2D$. Its eigenvalue degeneracies are the integer-zero multiplicities; its finite spectral parity is the Thue–Morse sequence; its zeta function is $\zeta(s)/(1-2^{-s})$;

its complex dimensions generate a log-periodic heat wave; and its power traces recover the probability cumulants.

The same operator admits two complementary q -constructions. The determinant $\mathcal{E}(u; Q) = \det(I - uQ^D)$ is a dyadic tower of q -Pochhammer symbols with exact finite Gaussian-binomial coefficients and explicit Mellin asymptotics. Replacing each spectral integer by its symmetric q -integer produces Φ_Q , an order-zero entire function with the same multiplicity and Thue–Morse sign word, an exact q -renormalization equation, and a locally uniform limit to the Rvachev Fourier transform. Formula [equation \(11.12\)](#) is the central new bridge: it connects q -Pochhammer symbols directly to the zero divisor of an infinite sinc product, while [equation \(11.7\)](#) shows that the connection is dictated by self-similarity rather than by a formal change of notation.

The resulting program is concrete. Its finite arithmetic core is ready for Lean formalization; its local jets provide stable numerical tools; its Mellin theory supplies exact asymptotics; and its unresolved comparison with the exact dyadic-value q -calculus offers a sharply defined next frontier.

A Recursive TeX corpus ledger

The following ledger records the recursive TeX corpus used for the audit. Directory prefixes are shortened relative to Analysis/FabiusFunction/docs/. For typographic wrapping, spaces in two historical directory names are displayed as underscores.

A.1 Living and canonical documents: 18

No.	Relative document
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6	non-formalized-research-frontiers/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
7	.../rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8	.../rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Rvachev_Up_Fourier_Decay.tex
9	.../Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10	.../Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11	.../Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12	.../asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13	.../rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14	.../rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15	.../rvachev_up_fourier_decay-5/rvachev_up_fourier_decay.tex
16	.../rvachev_up_fourier_decay-6/rvachev_up_fourier_decay.tex
17	.../rvachev_up_fourier_decay-7/rvachev_up_fourier_decay.tex
18	.../rvachev_up_fourier_decay-8/rvachev_up_fourier_decay.tex

A.2 Frozen historical archive: 32

No.	Archived document directory / TeX base name
1	early-notes/Fabius_Asymptotic/Fabius_Asymptotic.tex
2	early-notes/K-fold_summation_over_the_signed_Thue-Morse_sequence/K-fold_summation_over_the_signed_Thue-Morse_sequence.tex
3	standalone-studies/Non_Elementarity_of_the_Fabius_Function/...tex
4	standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
5	standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
6	primary-exposition/Fabius_Function_and_Rvachev_Up/...tex
7	primary-exposition/Fabius_Function_and_Rvachev_Up-2/...tex
8	primary-exposition/Fabius_Function_and_Rvachev_Up-3/...tex

No.	Archived document directory / TeX base name
9	primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
10	primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
11	primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
12	primary-supplements/Fabius_Function_Missing_Parts/...tex
13	primary-supplements/Fabius_Function_Missing_Parts-2/...tex
14	q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
15	q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
16	lean-walkthrough/Fabius_Function_Lean_Walkthrough/...tex
17	lean-walkthrough/fabius_lean_walkthrough/...tex
18	glossary/FabiusFunction_Mathematical_Glossary/...tex
19	glossary/FabiusFunction_Mathematical_Glossary-2/...tex
20	research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/...tex
21	research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/...tex
22	research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/...tex
23	research-frontiers/Fabius_Dyadic_q_Connections/...tex
24	research-frontiers/Fabius_Thue_Morse_Convergence_Rate/...tex
25	research-frontiers/Fabius_Thue_Morse_Convergence_Rates/...tex
26	research-frontiers/Primary_Exposition_Gap_Register/...tex
27	research-frontiers/Repeated_Integration_and_Rvachev_Up/...tex
28	research-frontiers/Rvachev_Up_from_Repeated_Integration/...tex
29	research-frontiers/non-formalized-research-frontiers/...tex
30	research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
31	research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
32	research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

A.3 Imported paper sources: 7

No.	Relative TeX source
1	papers/AIF/AIF_2017__67_2_637_0.tex
2	papers/AIF/AIF_2017__67_2_637_0-corrected.tex
3	papers/Lacunary_products/Lacunary_products.tex
4	papers/09-Function/09-Function.tex
5	papers/157-Arithmetic-v3/157-Arithmetic-v3.tex
6	papers/document/document.tex
7	papers/ubiq15/ubiq15-corrected.tex

B Residue calculations for the critical q -Euler product

For reference, the local series used in [theorem 10.2](#) are

$$\Gamma(s) = \frac{1}{s} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12} \right) s + O(s^2),$$

$$\begin{aligned}\zeta(1+s) &= \frac{1}{s} + \gamma - \gamma_1 s + O(s^2), \\ \frac{1}{1-e^{-Ls}} &= \frac{1}{Ls} + \frac{1}{2} + \frac{Ls}{12} + O(s^3), \\ \zeta(s) &= -\frac{1}{2} - \frac{\log(2\pi)}{2}s + \frac{\zeta''(0)}{2}s^2 + O(s^3).\end{aligned}$$

Hence

$$\begin{aligned}\Gamma(s)\zeta(1+s) &= \frac{1}{s^2} + \left(-\gamma_1 - \frac{\gamma^2}{2} + \frac{\pi^2}{12}\right) + O(s), \\ \frac{\zeta(s)}{1-e^{-Ls}} &= -\frac{1}{2L} \frac{1}{s} - \frac{1}{4} - \frac{\log(2\pi)}{2L} \\ &\quad + \left(\frac{\zeta''(0)}{2L} - \frac{\log(2\pi)}{4} - \frac{L}{24}\right)s + O(s^2).\end{aligned}$$

Multiplication by $t^{-s} = e^{sy}$ then yields the y^2 , y , and constant terms in equation (10.8).

C Selected source map

References

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